



Toward deep learning sequence–structure co-generation for protein design

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Deep generative models that learn from the distribution of natural protein sequences and structures may enable the design of new proteins with valuable functions. While the majority of today's models focus on generating either sequences or structures, emerging co-generation methods promise more accurate and controllable protein design, ideally achieved by modeling both modalities simultaneously. Here we review recent advances in deep generative models for protein design, with a particular focus on sequence-structure co-generation methods. We describe the key methodological and evaluation principles underlying these methods, highlight recent advances from the literature, and discuss opportunities for continued development of sequence-structure co-generation approaches.

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Introduction

Deep generative models trained on natural biomolecular diversity have the potential to generate valid and novel proteins to help solve modern-day challenges, for example as effective therapeutics or engineered enzymes [1–3]. Proteins are biomolecules comprised of a **sequence**¹ of amino acid **residues**, and this sequence dictates how a protein assembles into a 3-dimensional structure. Sequence and structure together mediate

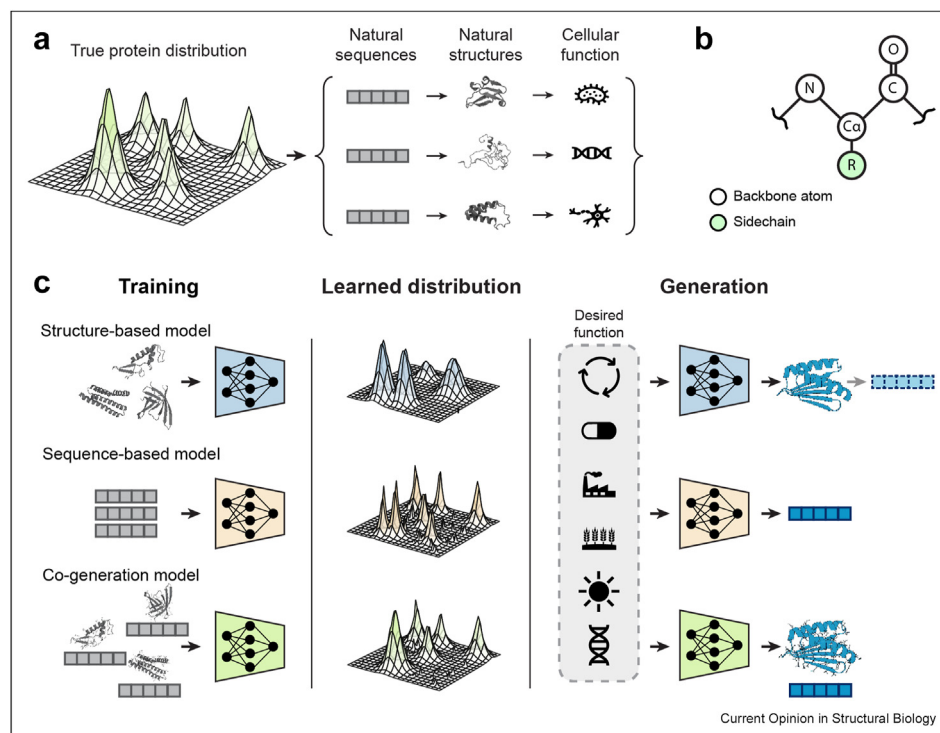
both natural and artificial protein function (Figure 1a). Impressively, only 20 natural amino acids are needed to describe the chemical and structural diversity of all natural proteins. Each of these amino acids share a set of common **backbone** atoms, differing only by the composition of their **side-chain** atoms. As a result, a protein's sequence can be easily described by a single string of single-letter amino acid codes. When available, structural characterization of proteins also provides 3D coordinates of the backbone and side-chain atoms that comprise their structures.

Because of the complexity of the protein sequence-structure landscape, computational methods have been developed to enable the engineering or *de novo* design of new proteins. Traditional physics-based protein design methods have successfully created a variety of functional proteins [4,5]; however, physics-based methods are constrained by the inaccuracies of empirical energy functions [6] and by their dependence on existing protein structures or predefined topologies [7]. In contrast, deep learning approaches that learn from large-scale protein sequence or structure databases enable the exploration of a vast and largely untapped design space, leading to the creation of proteins with novel sequences, folds, and desired functions [2,8,9]. This capability has opened up exciting possibilities in various applications such as designing novel enzymes with tailored catalytic activities [10,11], engineering high-affinity protein binders for therapeutic purposes [12,13], and even creating entirely new protein topologies with the potential for unprecedented functionality [14,15].

Deep generative modeling is a powerful approach in which a neural network is trained on samples from the distribution of natural protein sequences or structures in order to propose new ones. While the ultimate goal is to generate proteins conditioned on a specific function, practitioners often first train unconditional models and then repurpose or fine-tune models for conditional generation. Most current deep learning methods for protein design aim to generate functional proteins either by first generating a structural backbone and then designing a sequence that will fold into that structure (structure-based design), or by directly generating a

¹ Bolded terms are defined in Appendix A.

Figure 1



Generative modeling for protein design. (a) Natural proteins are (biased) samples from the distribution of possible sequences. Each protein's sequence determines its structure, and sequence and structure together mediate function. (b) The backbone atoms are common to all amino acid residues, while the identity of each residue is determined by the identity of the atoms in its side chain (R). (c) Overview of structure-based models, sequence-based models, and co-generation models for protein design.

protein sequence (sequence-based design). These methods have complementary strengths and weaknesses. Structure-based methods leverage more semantically rich information and are most effective when the desired function is mediated by a static structural **motif**. However, structural data is limited and biased toward a small subset of observed protein sequences, leading to less diverse learned distributions. Meanwhile, sequence-based methods generalize more easily to functions that rely on larger or multiple domains, are mediated by a structural ensemble, or involve intrinsically disordered regions, yet the lack of explicit structural constraints can lead to suboptimal generation quality. A deeper integration of the two modalities could significantly enhance the holistic understanding of protein properties. One modality often serves as a high-level abstraction of the other—for instance, a single protein sequence may represent an ensemble of structural conformations, while a single structure can be encoded by multiple distinct sequences. Bridging this gap is essential for developing models capable of capturing the full complexity of protein behavior.

To overcome the limitations of generating sequence or structure alone, researchers are developing sequence-

structure co-generation methods, which aim to model and generate both modalities simultaneously. Reasoning about sequence and structure throughout generation and fully leveraging the available data should result in a more accurate learned distribution and more plausible generations. This review aims to provide a comprehensive overview of recent advances in deep generative models for protein design, with a particular focus on sequence-structure co-generation methods. We begin by introducing recent advances in structure-based and sequence-based generation, laying the foundation for understanding the unique advantages and challenges of co-generation (Figure 1c). Throughout the review, we describe key principles, highlight useful examples from the literature, and discuss areas that would benefit from future methodological improvements.

Single-modality deep generative models of proteins

Structure-based methods

Structure-based protein design typically involves two stages: first generating a protein backbone structure and then finding a compatible amino acid sequence. This process can be represented as:

$$P(\text{seq}, \text{structure}) = P(\text{structure}) \times P(\text{seq}|\text{structure}). \quad (1)$$

Generative models of structure learn $P(\text{structure})$ and then rely on a separate **backbone-conditioned sequence design** model [16] to compute $P(\text{seq}|\text{structure})$. There are three notable approaches to modeling $P(\text{structure})$: hallucination,² denoising diffusion probabilistic models (DDPMs), and flow matching.

Hallucination methods [17–19] begin with a random sequence and use discrete optimization or back-propagation through a structure prediction model to find sequences that are predicted to fold to a stable structure with high confidence. Hallucinated sequences are often adversarial, so a high-quality sequence for the generated structure is redesigned using a backbone-conditioned sequence design method. Several recent hallucination approaches find that optimizing the “relaxed” sequence trajectory show superior performance on large monomer generation and binder design [20,21]. Denoising diffusion probabilistic models (DDPMs; [22,23]) are generative models trained to iteratively refine noised data into realistic protein backbones [24–27,15,12,28–30]. Flow matching models [31–36] generalize diffusion models by allowing mappings between arbitrary pairs of distributions through integration of ordinary differential equation (ODE) trajectories. This increased design space yields performance improvements. We provide a brief overview of diffusion and flow matching models in Appendix B.

The goal for structure-based methods is to produce designable backbones that are distinct from each other and from natural structures. **Designability** refers to the ability of current methods to find a sequence that is predicted to fold to the generated structure, and is estimated by predicting sequences from the generated backbone, passing those sequences to a structure prediction model, and measuring how self-consistent the generated and predicted structures are. Hallucination methods produce highly designable and diverse backbones without additional training; however, sampling from diffusion and flow matching models is much faster. Flow matching models claim superior generation quality over diffusion models while being more computationally efficient; however, there is a lack of rigorous ablation studies demonstrating the superiority of flow matching over diffusion models in protein structure generation. Structure-based models can be used to generate functional proteins via **motif scaffolding**, in which the structural information for a functional motif is used to conditionally generate a structural scaffold to hold the

motif in place [19,12], or via conditioning on text descriptions of function [15,37]. Structure-based models have demonstrated success across a range of tasks, including unconditional and topology-constrained monomer design, enzyme design, and binder design [12].

However, they are limited by the relatively small number of experimental structures (approximately 225,000 structures in the PDB [38]) available for training. Additionally, these models struggle to access functions mediated by disordered regions, which are underrepresented in the Protein Data Bank (PDB) as they rely on dynamics rather than a single stable structure [39]. Moreover, current structure generation methods fall short in modeling fine-grained structures that include side-chain conformations, which are critical for catalysis and protein–protein interactions as the identities and placement of the side-chain atoms are tied to both the amino acid types and backbone structure.

Sequence-based methods

Sequence-based methods seek to generate plausible proteins directly by training on the distribution of natural sequences. Protein sequence models are especially promising for functions that are not mediated by a single stable structure, those that do not fall explicitly into a binding-based paradigm, and those that involve intrinsically disordered regions. To date, the most notable approaches for sequence-based protein design are autoregressive and diffusion models. Autoregressive models predict each amino acid residue from the previous ones [40–45], while discrete diffusion methods are trained to iteratively convert corrupted sequences into realistic protein sequences [39,46,47]. In absorbing state, or masking, diffusion [48], residues are replaced by a special mask token, and the model is trained to predict the identities of the true residues given the unmasked residues [39,46,47]. Protein masked language models [47,49], which learn to reconstruct a fixed proportion of masked residues from the unmasked residues, can be considered a special case of absorbing state diffusion when used generatively. Discrete diffusion denoising probabilistic models are trained to iteratively denoise a sequence that has been randomly mutated [50,39]. Continuous diffusion models can also be applied to generate protein sequences by operating in the latent space of another pretrained model [51,52].

For unconditional generation, autoregressive models have been shown to outperform masking and discrete diffusion models at learning the sequence distribution [39,53]. However, masking diffusion models have demonstrated the ability to perform motif scaffolding directly in sequence space [39,46]. Sequence-based models have also been used to generate sequences with a specified function by conditioning on the desired structure [14], fine-tuning on a family of natural proteins with the desired function [54,40,55], conditioning

² While hallucination methods generate both sequence and structure, they utilize only the generated structure and do not directly model $P(\text{structure})$, instead using structure prediction and backbone-conditioned sequence design models to match sequences to generated structures.

on the desired reaction class [56,57], and using gradients from a function prediction model [52,58].

Compared with structure-based methods, sequence-based methods have observed a much broader range of proteins: UniProt contains about 300 million sequences [59]. However, despite accessing this extensive sequence space, these models often fail to generate truly *de novo* proteins. Functional sequences generated from sequence-based generative models are often chimeras of several natural proteins instead of being truly *de novo* [55,47], although it is possible to sample stable examples of novel structures from sequence-based models [14]. While they are aware of residue identities and therefore implicitly learn side-chain distributions and relationships, current sequence generation methods do not reason directly about the placement of side-chain atoms.

Sequence-structure co-generation

Sequence–structure co-generation aims to combine the strengths of structure- and sequence-based modeling and to unlock atomistic control over protein design. Rosetta’s “flexible backbone design” protocol was one of the earliest to adopt this philosophy [60], by starting with a coarse-grained designed backbone and then alternating between designing a sequence given the current backbone and optimizing the backbone given the current sequence. While this approach has been successful in applications such as designing small protein binders [4], it relies on hand-crafted energy functions for both backbone and sequence design instead of learning the underlying distributions from data; in fact, using neural network models for structure prediction or backbone-conditioned sequence design in this pipeline greatly improved success rates [13]. However, the flexible backbone design approach only allows small-scale adjustments to the initial structure, limiting the extent to which sequence and backbone compatibility can be optimized. An early attempt at co-generation using deep learning focused on “inpainting” tasks, in which a region of a protein is masked out and regenerated jointly with both structure and sequence [19]. However, the true potential of co-generation lies in generating entire proteins. Recent advances have enabled models to generate both sequence and structure from scratch. The central challenge of co-generation is how to pass information between the sequence and the structure. Current strategies for mixing sequence and structure information within the model include tokenizing the structure, leveraging a structure prediction model, mixing discrete and continuous processes, and all-atom generation (Figure 2).

Token-based co-generation models

One strategy for co-generation is to first convert the continuous structure to a sequence of discrete structure

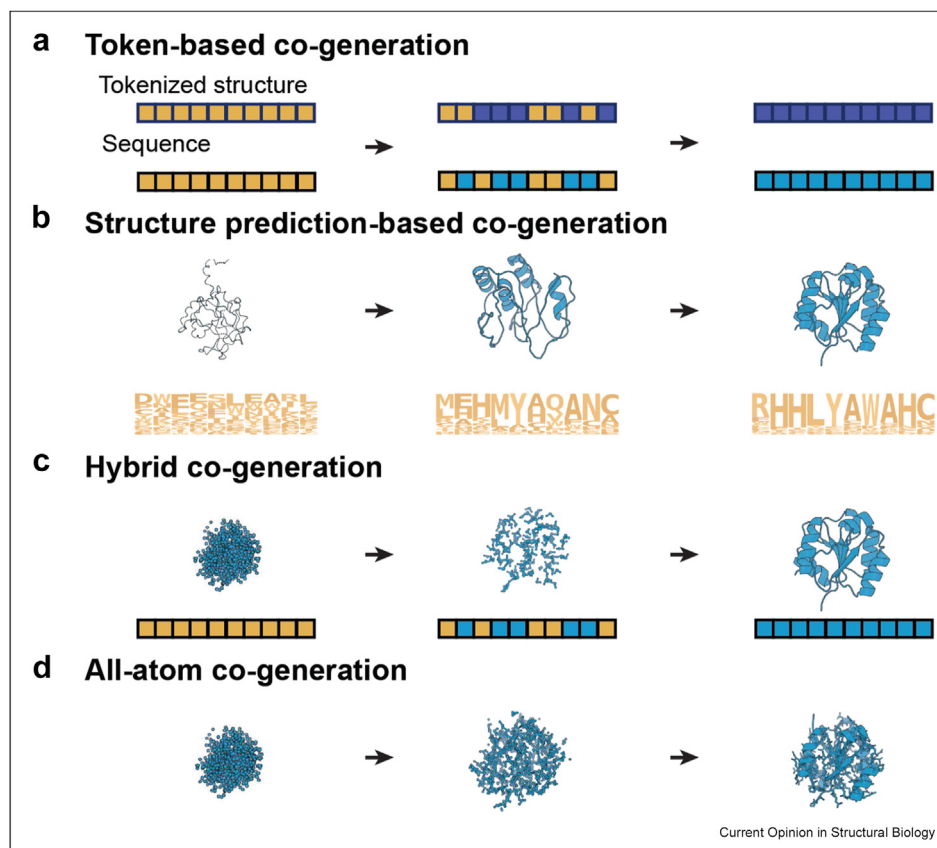
tokens using techniques such as vector-quantized variational autoencoders (VQ-VAEs) [61] (Figure 2A). Concatenating the sequence and structure tokens then enables language-model style architectures to be repurposed for co-generation. Foldseek [62] applied VQ-VAEs to tokenize the structural environment around each residue in order to use efficient sequence-based homology search methods for structural homology search and clustering. This tokenization scheme was later applied to masked language models to train joint representations of sequence and structure [63,64]. In ESM3 [47], the authors train their own structure tokenizer and show that their multimodal architecture enables the unconditional generation of high-quality proteins, motif scaffolding, and, by incorporating function tokens, function conditioning. Similarly, DPLM-2 [65] incorporates discrete structural tokens into the pretrained DPLM framework to enable co-generation, further emphasizing the growing interest in token-based strategies for sequence–structure modeling.

Co-generation methods that tokenize structure can take advantage of the extensive neural network machinery for modeling sequences of tokens and can train on datasets where not every sequence is paired with a structure. However, they do not directly model side-chain atom positions during generation, and instead use the VQ-VAE decoder to convert structure tokens to all-atom coordinates after generation. Furthermore, while previous work has shown that autoregressive language models outperform masked language models for generation, there is not yet an autoregressive joint sequence–structure token model. Finally, the quality of these models is upper-bounded by the fidelity and expressiveness of the structure tokenization, and the design choices here remain mostly unexplored.

Structure prediction-based co-generation models

Instead of discretizing structural information, ProteinGenerator [66] employs a diffusion process in continuous sequence space to generate protein sequence and structure jointly. By leveraging the pretrained RoseTTaFold2 network [67] for structure prediction, the model predicts and conditions on the corresponding structure at each diffusion step (Figure 2b). This approach circumvents the need to explicitly model the structure distribution. ProteinGenerator has shown experimentally validated ability to scaffold flexible peptides and to design proteins capable of adopting multiple conformations, a challenging task that requires simultaneous optimization on distinct conformational states. Unlike previous methods that need predefined desired conformations before sequence optimization [68], ProteinGenerator achieves this by tying the sequences from several generation processes together with distinct secondary structure guidance. While theoretically applicable to a broader range of design

Figure 2



Overview of current sequence–structure co-generation methods. (a) Token-based co-generation models convert structure to discrete tokens. (b) Structure prediction-based co-generation models make sequence continuous and guide generation with a structure prediction model. (c) Hybrid co-generation models use the same neural network to generate a discrete sequence and a continuous backbone structure. (d) All-atom co-generation models directly generate the identity and location of every atom in the structure.

tasks, ProteinGenerator lags behind RFDiffusion in motif scaffolding and unconditional generation of larger proteins, suggesting limitations associated with the absence of explicit training on structure generation. However, generation in sequence space may enable the use of sequence-based function predictions for conditioning.

Hybrid co-generation models

Hybrid co-generation models combine the discrete sequence-generation process with the continuous backbone generation process without altering the fundamental properties of the underlying data (Figure 2C). CarbonNovo interleaves, at every time step, a diffusion model for backbone structure generation with a structure-conditioned Markov random field for sequence generation [69]. In contrast, DiffAb [70] and MultiFlow [71] add a discrete sequence diffusion or flow matching process to a diffusion- or flow matching-based backbone generation model. These methods use the same neural network and a shared representation to learn two intertwined generative processes.

These hybrid models simply and efficiently combine advances in discrete and continuous generative models without dealing with the complexity and possible loss in fidelity of tokenizing structure or embedding sequence. However, they require paired sequence-structure data and do not directly model side-chain atom positions during generation. The suitability of simply incorporating sequence generation tasks into existing model architectures for backbone generation is an important question that warrants further exploration. Current hybrid co-generation models have yet to demonstrate substantial gains in designability and diversity over RFDiffusion. Besides motif scaffolding, function conditioning has yet to be explored with these methods, although the inclusion of sequence information should enable the use of sequence-based function predictors for conditioning.

All-atom co-generation models

Direct generation of all-atom protein structures (i.e. the complete set of protein backbone and side-chain atoms) implicitly enables co-generation as the side-chain atoms

directly represent the amino acid identities that make up a protein's sequence (Figure 2d). The central challenge in all-atom co-generation lies in the model's requirement to flexibly handle a varying number and type of side-chain atoms at each step. At the initial sampling stage, the backbone, sequence, and number of atoms are all unknown. Protpardelle [72] employs Euclidean diffusion for both backbone and side-chain atoms. To circumvent the variable-residue challenge, Protpardelle tracks all 73 possible non-hydrogen atoms for each residue. At each diffusion step, the predicted backbone guides sequence design, followed by the denoising of corresponding side-chain atoms in the atom73 coordinates. To bypass the need for decoding the amino acid identity in an intermediate step, a more recent work, P(all-atom) [73], proposes atom14 coordinates as a unified atomic representation as the largest amino acid has 14 non-hydrogen atoms. The model then applies simple Euclidean diffusion on these atom14 coordinates without intermediate sequence decoding. Because each amino acid type has a distinct collection of side-chain conformations, the model learns to implicitly assign an amino acid identity to each residue, based on specific conformational patterns within the atom14 coordinates. The sequence is thus decoded after generation, and the generated atom14 coordinates are mapped back to the all-atom atomic index for the assigned amino acid.

All-atom generation models hold great promise for protein design as they promise precise control over side-chain specific interactions throughout the generative process. This level of control is critical for tackling challenging tasks such as enzyme and antibody design. All-atom generation naturally extends to the co-generation of non-peptide interacting molecules, such as DNA, RNA, or small molecules. However, the high dimensionality of all-atom representations poses significant computational hurdles, necessitating the development of more efficient model architectures [74]. In addition, it is unclear how to leverage sequences without structures in an all-atom model. Overall, this remains a largely unexplored but fertile area for future research.

Discussion and outlook

This review examines the early efforts at using deep learning to co-generate protein sequence and structure. While these efforts have yielded promising results, there is still room to improve the consistency of the generations, to fully leverage unpaired sequences for training and evaluation, and to test the capabilities of co-generation for real-world applications.

Self-consistency between the generated sequence and structure remains challenging. Current techniques model sequence and structure jointly but ultimately sample them independently given the model outputs.

Furthermore, some methods use an absorbing state diffusion process for the sequence without iterative refinement—once unmasked, an amino acid identity cannot be corrected as the generated structure is refined. Therefore, current co-generation models produce less designable structures than two-stage structure-based methods. For example, co-generated sequences from Protpardelle are less self-consistent with a generated backbone structure than sequences designed from the same structure using ProteinMPNN [72]. By training on the most self-consistent sequences generated by ProteinMPNN from native and generated backbones, Multiflow's [71] co-generation method achieves comparable self-consistency to an equivalent two-stage process. While ESM3 [47] theoretically allows simultaneous generation of sequence and structure, its green fluorescent protein (GFP) design pipeline alternates between generating structure tokens based on functional motifs and then optimizing the sequence based on the structure, repeated for multiple cycles.

While generative models that unify both modalities would ideally combine the strengths of structure- and sequence-based methods, current approaches often overemphasize structure in their architecture, data, and evaluations, potentially limiting their ability to explore the full design space where a static structure is insufficient to determine function—crucial for tasks like conformation switching and enzyme design. Many co-generation methods are either dependent on a pretrained structure prediction module or are direct adaptations of architectures designed for backbone generation. All the co-generation methods we cover require paired sequence-structure data. Methods that use structure when available but default to sequence-only when not remain underexplored and thus the scope of available sequence data has not been fully exploited for co-generation. While using *in silico*-predicted structures seems like a possible way to bridge this gap, naive augmentation has widely been observed to harm generation quality due to overfitting [64,36]. Alternatively, a recent model that generates in the latent space of a structure prediction model produces all-atom structures without using any structures during training [75].

This bias toward structure is perhaps most apparent in our reliance on structural self-consistency for evaluating generation quality and in the dominance of structure-focused tasks, such as binder design or motif scaffolding, for downstream applications. These evaluations favor sequences that fold into stable structures (or, conversely, very stable backbones), limiting exploration of applications where function-driven dynamics are more important than rigidity. Self-consistency metrics are inherently constrained by the capabilities of current structure prediction and backbone-conditioned sequence design models. The structure prediction models commonly used in these pipelines can be

insensitive to the effects of sequence variation that are not seen in nature. For example, for generated enzymes, AlphaFold2 prediction confidence does not correlate with activity or expression [76]. In addition, fewer than half of the high-confidence structures in AlphaFoldDB redesigned by ProteinMPNN and predicted by AlphaFold2 achieves root mean square deviation (RMSD) $< 2\text{\AA}$ [30]. These observations suggest that predicting sequence from backbone is easier for some proteins, while predicting structure from sequence is easier for others. This property motivates the development of performant co-generation models that can learn from both sequence and structure, simultaneously. By generating both in tandem, along with modalities like protein surface properties, features inherently linked to both sequence and structure [77], methods for protein sequence–structure co-generation could overcome the complexities of mapping between these individual

modalities and ultimately enable a broader, more diverse range of downstream applications for generative protein design.

Declaration of competing interest

The authors declare that they have no known competing financial interests or personal relationships that could have appeared to influence the work reported in this paper.

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Appendix A. Glossary

Table A.1

Glossary of terms used in the review.

Term	Description
All-atom co-generation	A co-generation strategy that directly generates the complete protein structure, including all backbone and side-chain atoms.
Backbone	A protein's backbone consists of its N, C α , C, and O atoms. All amino acids contain these atoms, so the backbone does not depend on the identities of the amino acids.
Backbone-conditioned sequence design	The design of an amino acid sequence that is likely to fold into a predefined three-dimensional protein backbone.
Designability	The degree to which a backbone structure can be realized by an amino acid sequence.
Diversity	The degree to which generated structures or sequences differ from each other.
Hybrid co-generation	A co-generation strategy where the protein structure is represented in its native continuous form and the protein sequence is represented in its native discrete form.
Motif	A specific region in a protein with a conserved sequence or structural pattern, often associated with a specific function.
Motif scaffolding	The design of a new protein that incorporates a specific motif as a core element, with the surrounding protein context designed to support and enhance the motif's function.
Novelty	The degree to which generated structures or sequences differ from naturally occurring proteins.
Residue	An individual amino acid within a protein.
Self-consistency	The agreement between a generated sequence and its corresponding generated structure. A self-consistent sequence is likely to fold into the intended structure, typically assessed using structure prediction tools.
Sequence	The linear order of amino acid residues in a polypeptide chain, typically listed from the N-terminal to the C-terminal.
Side chain	The variable chemical group (R) attached to the alpha carbon of an amino acid, determining its identity and chemical properties.
Structure prediction-based co-generation	A co-generation strategy that leverages a structure prediction network to jointly generate the protein sequence and structure, where the structure is predicted from the sequence.
Token-based co-generation	A co-generation strategy where the protein structure is converted into discrete tokens, enabling the use of language model-like architectures.

Appendix B. Diffusion models and flow matching

Flow matching and diffusion models are the state-of-the-art methods for generating data in continuous spaces, such as Euclidean spaces and manifolds [35]. Their goal is to generate realistic samples that resemble the training dataset. These techniques are used in fields ranging from text-to-image, text-to-audio, and text-to-video models, to the modeling of protein backbones. In simple terms, these methods create new data by transforming random noise into structured data that matches the distribution of the data of interest.

Training flow matching models involves two main objects: the reference flow and the generative flow. The reference flow is constructed by drawing random variables from an initial distribution, $\bar{X}_0 \sim p_0$, which represents random noise, and from the distribution of the data that we want to model, $\bar{X}_1 \sim p_1 = p_{\text{data}}$. In practice, $\bar{X}_1$ is sampled from the training dataset. Then, the reference flow is defined as an interpolation path $\bar{X}_t$ between the noise sample $\bar{X}_0$ and the data sample $\bar{X}_1$. For example, in Euclidean space, it is the standard to choose p_0 to be the multivariate standard Gaussian, and a linear interpolation path $\bar{X}_t = \beta_t \bar{X}_0 + \alpha_t \bar{X}_1$ for t between 0 and 1, where the functions α_t and β_t are chosen appropriately.

The generative flow X_t is a process that begins from the same noise as the reference flow ($X_0 \sim p_0$) and evolves to produce new samples that look like the real data. The key idea is that, at any time t , the distributions of the generative flow X_t and the reference flow $\bar{X}_t$ are identical, so by the final time $t = 1$, X_1 follows the same distribution as the data. In flow matching, the generative process follows an ODE defined as:

$$\frac{dX_t}{dt} = v_\theta(X_t, t), \quad X_0 \sim p_0, \quad (\text{B.1})$$

Here, $v_\theta(\cdot, \cdot)$ is a velocity field, which is usually modeled using neural networks, such as a U-Net [78] or diffusion transformer architecture [79]. The goal is to optimize the network so that it matches the behavior of the reference flow over time, which is achieved by minimizing the difference between the predicted velocity and the actual derivative of the reference flow [31,33]. Once v_θ has been trained, generating an artificial sample involves drawing a random variable from p_0 and solving the ODE (B.1) using the numerical method of choice, such as the Euler scheme.

Diffusion models [22,23] are a specific form of flow matching on Euclidean spaces, where the reference flow uses a multivariate standard Gaussian as the noise distribution, and linear interpolation paths with coefficients $\alpha_t = \sqrt{\bar{\alpha}_t}$ and $\beta_t = \sqrt{1 - \bar{\alpha}_t}$. In this case, instead of directly learning the velocity field, the neural network is a noise predictor ϵ_θ that predicts the original

noise X_0 . The training objective is to minimize the error in this noise prediction. Diffusion models appeared before flow matching, but they are being superseded by the latter due to improvements in sample generation quality and efficiency.

Data availability

No data was used for the research described in the article.

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Papers of particular interest, published within the period of review, have been highlighted as:

* of special interest

** of outstanding interest

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